XRPL Validator — Daily Engineering Report · 2026-10-18

Period: 2026-10-18 00:00 to 24:00 US Eastern (2026-10-18 04:00 to 2026-10-19 04:00 UTC); ledgers #107,731,616 to #107,753,847. Every count from the ledger and the validator's logs stops at #107,753,847, closed 03:59:51 UTC. Generated 2026-10-21 00:48 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records; Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores. Not available: none.

Status

Impaired (all causes resolved or fixed) — 12 validations of a ledger hash the network did not validate.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,213 of 22,232 ledgers; 12 validations did NOT match the network's validated ledger. Unsigned: 19 during planned restarts.
- **Native engine:** 0 engine receipts recorded.
- **Changes:** 2 deployments (cycles 182, 183), 2 commits merged into main.
- **Amendment fixCleanup3_4_0** holds majority; activation due ~2026-10-23 09:34 UTC.

Service levels

Measure	Value
Ledgers closed by the network	22,232
Validations issued	22,213
Coverage (validations / ledgers closed)	99.915%
Validations matching the network's validated ledger hash	22,201 of 22,213
Unsigned during planned restarts	19
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	2

Measure	Value
Basis: the hash the first peer announced as accepted (before cycle 183)	16,883
Basis: the hash the trusted quorum named	5,330
External agreement, data.xrpl.org (UTC day 2026-10-18)	0.99550 (83 missed of 18,450)

Restarts:

- 15:29:18 UTC: deployment of cycle 182; validations resumed at #107,742,281 (10 ledgers unsigned between the last validation before the restart and the first after it)
- 22:17:48 UTC: deployment of cycle 183; validations resumed at #107,748,518 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,731,616–#107,753,847), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipts this day.

Incidents

INC-01 · 12 validations of a ledger hash the network did not validate

Window	2026-10-18 07:58 UTC to 2026-10-18 21:06 UTC
Ledgers	#107,735,345, #107,744,188, #107,744,609, #107,744,703, #107,744,972, #107,745,580, #107,745,624, #107,746,244, #107,746,300, #107,746,977, #107,747,014, #107,747,426
Impact	the validator signed a hash the network did not validate for these ledgers; data.xrpl.org scores such ledgers missed
Cause	Before cycle 183 the validator signed the ledger hash from the first peer that announced a ledger as accepted, without confirmation. For these ledgers that peer had built a different ledger, so the validator signed a hash the network never validated.

Action	commit d540e6e — fix(validator): sign a ledger only once the trusted validators reach quorum on its hash; deployed in cycle 183 at 2026-10-18 22:18 UTC

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
182	`a6ab3ee`	15:30:08 UTC	2,159 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	408 matched, 0 mismatched	#57 from 15:56 UTC: ended early: superseded by the next deployment (2026-10-18 22:18 UTC)
183	`d540e6e`	22:18:38 UTC	2,161 passed / 0 failed; port 1,006 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	406 matched, 0 mismatched	#58 from 22:45 UTC: CLEAN (2026-10-19 22:52 UTC)

Cycle 182 delivered: `a6ab3ee` fix(peer): our validations get their own queue and go to every regular peer; peers' transactions are no longer relayed to the hubs

Cycle 183 delivered: `d540e6e` fix(validator): sign a ledger only once the trusted validators reach quorum on its hash

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
14:45 UTC	xrpl-validator	`a6ab3ee`	fix(peer): our validations get their own queue and go to every regular peer; peers' transactions are no longer relayed to the hubs
21:37 UTC	xrpl-validator	`d540e6e`	fix(validator): sign a ledger only once the trusted validators reach quorum on its hash

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(peer): our validations get their own queue and go to every regular peer; peers' transactions are no longer relayed to the hubs** (`a6ab3ee` , xrpl-validator) data.xrpl.org counted 53 of 19,441 ledgers missed in a day, all of them signed. Two ways a signed validation was lost: - It shared a 2,000-place queue to the hubs with every transaction a peer sent us. A ledger of 1,000-2,000 transactions overflowed it, and a session that fell behind skipped the oldest frames, our validation among them (#107741340: four hubs connected, three public servers never saw it). - It went …
- **fix(validator): sign a ledger only once the trusted validators reach quorum on its hash** (`d540e6e` , xrpl-validator) Until now the validator signed the hash the first peer announced as accepted. On mainnet #107746977

and #107747014 (18 Oct) that peer had built another ledger, so we signed a hash the network never validated, and data.xrpl.org scored both ledgers missed.

Independence program

No independence record for UTC day 2026-10-18; records begin 2026-10-20.

Network

Measure	Value
Ledgers closed	22,232 (#107,731,616–#107,753,847)
Mean close interval	3.89 s
Longest close interval	10 s (before #107,753,316)
Closes at least 15 s apart	0
Empty ledgers	0

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Open items

- Amendment fixCleanup3_4_0: activation due ~2026-10-23 09:34 UTC.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
10-18 14:42	10-18 14:45	xrpl-validator	`a6ab3ee`	fix(peer): our validations get their own queue and go to every regular peer; peers' transactions are no longer relayed to the hubs

Committed	Merged into main	Repo	Commit	Subject
10-18 21:33	10-18 21:37	xrpl-validator	`d540e6e`	fix(validator): sign a ledger only once the trusted validators reach quorum on its hash

Times in UTC. A commit appears on the day it was committed to main or merged into it.